Cloé Brenna, PhD

Computational Biologist | Machine Learning | Data Analysis

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Professional Summary

Computational biologist with a PhD from UNIL-CHUV, specialized in integrating spatial transcriptomics, multiplex imaging, and machine learning to analyze complex biological systems. Doctoral work focused on applying clustering, classification, and dimensionality reduction to high-dimensional datasets, combined with reproducible Python/R pipeline development. Engineering background in biotechnology and e-health provides strong foundation in data analysis, algorithm design, and cross-functional collaboration. Proven ability to translate complex biological questions into data-driven analytical frameworks, develop robust ML workflows, and deliver actionable insights. Seeking to apply expertise in machine learning, multi-modal data integration, and collaborative R&D to solve real-world challenges in industry settings.

Education

PhD in Life Sciences, Immunology & Computational Biology

UNIL-CHUV, Lausanne, Switzerland | 2022 – 2026

Thesis: "Delineation of lymphoid organ immune landscaping in health and viral infections: the role of aging." Integration of spatial transcriptomics, multiplex imaging, and machine learning.

Engineering Degree in Biotechnology & E-Health (M.Sc. level)

ESIEE Paris, France | 2019 – 2021

Specialization in computational biology, data analysis, and health technologies.

Bachelor's Degree in Biology, Health & Life Sciences

UPEC, Paris-Créteil, France | 2016 – 2019

Professional Experience

PhD Researcher – Computational Biology & Machine Learning

Institute of Pathology, CHUV / UNIL, Lausanne | 2022 – 2026

- Designed ML workflows (clustering, classification, dimensionality reduction) for spatial transcriptomics (Visium, GeoMx) and multiplex imaging at single-cell resolution.
- Developed reproducible Python/R pipelines for data processing, feature extraction, and model training enabling cell phenotype identification and spatial analysis.
- Integrated spatial transcriptomics + mIHC data to study immune landscapes in COVID-19 and aging using data harmonization and batch correction.
- Built predictive models (regularized regression, random forests) correlating molecular signatures with clinical outcomes.
- Collaborated with biologists, pathologists, and data scientists; mentored junior researchers in Git and ML best practices.

Research Intern – Machine Learning for Embedded Systems

Techno-Pôle, HES-SO Valais-Wallis, Sierre, Switzerland | Feb 2021 – Jul 2021

- Improved and tested real-time KNN-based sEMG classifier on Raspberry Pi for prosthetic control; optimized feature extraction and inference latency.
- Utilized Python, scikit-learn, and MATLAB for algorithm prototyping and performance benchmarking.
- Collaborated with hardware engineers to integrate ML model into end-to-end system.

Research Intern – Translational Immunology & Flow Cytometry Data Analysis

INSERM Cochin Institute, Paris, France | May 2020 – Jul 2020

- Applied flow cytometry analysis and statistical modeling to investigate oxidative stress modulation in systemic scleroderma.
- Used R/Bioconductor for differential expression analysis integrating clinical and experimental data.

Research Intern – Inflammation & Wound Healing Biology

INSERM Cochin Institute, Paris, France | May 2019 – Jun 2019

- Studied microbiota and olfactory receptor roles in wound healing using molecular biology techniques.

Machine Learning & Computational Expertise

ML Methods – Applied to Biological & High-Dimensional Data

Supervised Learning: Classification, Regression, Predictive Modeling
Unsupervised Learning: Clustering (K-means, hierarchical), Dimensionality Reduction (PCA, UMAP, t-SNE)
Model Validation: Cross-validation, Hyperparameter Tuning, Train/Test Splits
Interpretability: Feature Selection, SHAP values, Feature Importance Analysis
Regularization: Ridge, Lasso, Elastic Net
Performance Metrics: Accuracy, Precision, Recall, F1, RMSE, R^2

Experienced in building and validating ML models for cell phenotyping, biomarker discovery, and outcome prediction. Applied to spatial transcriptomics, single-cell RNA-seq, and multiplex imaging datasets.

Python Ecosystem – End-to-End ML & Data Science

Core: scikit-learn, pandas, numpy, scipy, statsmodels
Deep Learning: PyTorch, TensorFlow, Keras
Bioinformatics: scanpy, anndata, umap-learn
Visualization: matplotlib, seaborn, plotly
Image Analysis: scikit-image, opencv

Full-stack ML proficiency: data loading & preprocessing, exploratory analysis, model development (classical Machine Learning), and visualization. Built reproducible pipelines for large-scale biological datasets.

R / Bioconductor – Genomics & Spatial Analytics

Single-cell & Spatial: Seurat, SingleCellExperiment
Differential Expression: DESeq2, edgeR, limma
Pathway Analysis: clusterProfiler, fgsea
Visualization & Integration: ggplot2, tidyverse, harmony, combat
Reporting: shiny, knitr

Specialist in single-cell and spatial transcriptomics analysis, differential expression, pathway enrichment, and batch effect correction. Proven ability to deliver publication-ready analyses.

Reproducible Workflows & Development Tools

Version Control: Git/GitHub
Development: Python/R scripting, Jupyter, RStudio
Engineering: Pipeline development, Data QC & Harmonization
Standards: FAIR data principles, LaTeX, Scientific writing

Version control, modular pipeline design, and documentation aligned with FAIR principles. Experience in collaborative development and ensuring reproducibility across projects.

Bioinformatics & Multi-Omics Analysis

Spatial Transcriptomics: Visium, GeoMx – full analysis pipeline
Single-cell: RNA-seq analysis, cell type annotation & clustering, cell deconvolution
Integration: Multi-omics integration (spatial + mIHC), batch correction & normalization
Functional Analysis: Differential gene expression, pathway enrichment (Reactome, KEGG, GO)

End-to-end workflows from raw data to biological interpretation: QC, normalization, integration of multi-modal datasets, and spatial pattern visualization. Proven expertise in extracting actionable insights from complex omics data.

Imaging Analysis & Laboratory Techniques

Imaging: Multiplex immunofluorescence (Akoya/Opal), confocal & multiplex imaging (Leica, Polaris, Thunder), image segmentation, quantitative analysis, spatial interaction & nearest-neighbor analysis
Wet Lab: Flow cytometry / Histocytometry, RNAscope, tissue processing (FFPE), histology, antibody panel design

Additional Expertise & Continuous Learning

- **Cross-functional Collaboration:** Worked with biologists, pathologists, and data scientists to translate biological questions into analytical frameworks.
- **Reproducible Research:** Developed version-controlled, documented pipelines ensuring reproducibility and knowledge transfer.
- **Scientific Communication:** Presented at national conferences (Swiss Society of Pathology, DM-DMLP) and authored first-author publications.
- **Mentorship:** Guided junior researchers in computational methods, coding best practices, and ML workflows.
- **Continuous Learning:** Expanded proficiency in deep learning (PyTorch, TensorFlow) and advanced spatial analysis during doctoral research.

Selected Projects & Methodologies

- **Spatial Data Pipelines:** Built reproducible Python/R workflows for Visium/GeoMx data: QC, normalization, clustering, visualization.
- **Multi-modal Integration:** Combined spatial transcriptomics with multiplex immunofluorescence to characterize immune landscapes in COVID-19 and aging.
- **ML for Cell Phenotyping:** Applied clustering, classification, and dimensionality reduction to identify cell populations from high-dimensional imaging/transcriptomics data.
- **Data Quality Improvement:** Led quality assessment for spatial datasets in collaboration with pathologists and biologists.

Selected Publications

Brenna C, Bramon Mora B, Ioannidou K, et al. Bcl6 expression is associated with a distinct immune landscape and spatial transcriptome in COVID-19. *JCI Insight*, 2025.

Brenna C, Petrovas C. Germinal center immune dynamics: challenges for effective vaccination in the elderly. *Immunotherapy*, 2025.

Chantziou A, **Brenna C**, Ioannidou K, et al. HIV infection is associated with compromised tumor microenvironment adaptive immune reactivity in Hodgkin lymphoma. *Blood Adv*, 2025.

Burgermeister S, Orfanakis M, Georgakis S, **Brenna C**, et al. Unsupervised Clustering of Cell Populations in Germinal Centers Using Multiplexed Immunofluorescence. *Biology (Basel)*, 2025.

Georgakis S, Orfanakis M, Fenwick C, **Brenna C**, et al. Delineation of the Human Germinal Centre Immune Landscape Using Multiplex Imaging Analysis. *Immunology*, 2025.

Georgakis S, Ioannidou K, Bramon Mora B, **Brenna C**, et al. Cellular and molecular determinants mediating the dysregulated germinal center immune dynamics in systemic lupus erythematosus. *Front Immunol*, 2025.

Communications

- **May 2023 – DM-DMLP Research Day:** Oral presentation on spatial transcriptomics in COVID-19.
- **Nov 2023 – Swiss Society of Pathology Congress:** Poster on immune landscaping and spatial transcriptomics.

Languages

French: Native
English: C1 (Full professional proficiency)

Interests

Dodgeball, Martial Arts, Running, Biking, Salsa, Theater, Drawing, Hiking

References

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